Venkata Lakshmi Parimala Pasupuleti

Software Engineer · Python · AI Systems · Agentic Pipelines · Production Scale · AWS · Microservices · LangGraph · FHIR · HUD/LIHTC

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PROFESSIONAL SUMMARY

Software engineer with 8 years building production-grade systems for property managers — the exact workflows EliseAI's housing AI sits on top of. Shipping **20+ modules end-to-end** (1M+ records, 100K+ MAU, 99.5% uptime, reliability at scale) including Entrata's first production AI pipeline on 1.6M real records; led rollout of Docker microservices (50% speedup), AWS migration (100% PHI compliance), and startup-style 0→1 tooling implementation with measurable ROI across cross-functional stakeholder teams — bias for action, measurable impact every sprint. Built **Leasel** (agentic RAG housing compliance, 93.6% precision), **AgentProbe** (multi-agent LLM experimentation & eval framework, LangGraph, machine learning evaluation, continuous integration via GitHub Actions), and **AI on FHIR** (HIPAA-compliant NLP over medical records, FHIR APIs). M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

WORK EXPERIENCE

George Mason University
Aug 2024 – Aug 2025

Research Associate — ML Pipelines · AI Automation · Stakeholder Reporting

- Built end-to-end ML pipelines (Python, R, 190+ countries) achieving **$R^2 = 0.977$**; applied Isolation Forest anomaly detection across 3 research pipelines; automated statistical reporting with Python + Quarto driving **40% improvement** in stakeholder comprehension and adoption of research outputs.

Entrata
Jan 2021 – Aug 2024

Senior Software Engineer — Production AI · AWS · Microservices · System Design · Cross-functional Delivery

- Owned **20+ production-grade modules end-to-end** (Python, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints with cross-functional stakeholder teams — **1M+ records, 100K+ MAU, 99.5% uptime**; led architecture, design patterns, implementation, rollout, and adoption tracking for every feature shipped.
- Shipped Entrata's **first production AI pipeline on 1.6M real financial records** (PyTorch, OpenAI API, quality gates); led Docker microservices rollout (50% reliability gain); migrated to **AWS S3/RDS/Glue** (100% PHI compliance, zero incidents); optimized 200+ SQL queries for 30% performance improvement.
- Built **HUD and LIHTC (Section 42) compliance tooling** — income eligibility, AMI thresholds, Section 8/HCV certification, tax credit recertification pipelines across 100K+ unit portfolios; eliminated **80% of manual workload** with measurable ROI and zero compliance violations.

Entrata
Jan 2019 – Dec 2020

Software Engineer — HUD/LIHTC Compliance · Python · Data Pipelines · API Integrations

- Built HUD and LIHTC tax credit compliance automation processing **millions of tenant certifications** — Section 42 income limits, household composition rules, asset verification thresholds encoded into deterministic systems with full audit trails; built **API integrations and webhooks** across property management systems; conducted 300+ peer code reviews.

Entrata
Jan 2017 – Dec 2018

Associate Engineer — Full-Stack · Python · PostgreSQL · REST APIs · Prototyping

- Built and shipped full-stack features and SQL data layers (PHP, JavaScript, PostgreSQL) for property management workflows; rapid prototyping of new reporting tooling from 0→1 through production rollout across 100K+ unit portfolios.

PROJECTS

Leasel: Agentic Housing Compliance Pipeline
HOUSING AGENTIC RAG

Python · LangGraph · Deterministic Rules Engine · LLM-as-Judge · Pass@k Eval — github.com/venkatalakshmiparimala/Lease_complaince_rag

- **93.6% precision, 87.9% F1** across 10 housing compliance rules on 1,240 lease PDFs — LangGraph agentic RAG pipeline, 5 US states + federal standards, deterministic rules engine for full auditability; pass@k eval harness (pass@1 81.7%, pass@3 88.3%).

AgentProbe: Multi-Agent AI Experimentation & Eval Toolkit
MULTI-AGENT LLMOPS

Python · LangGraph · FastAPI · ChromaDB · LLM-as-Judge · OpenAI · Anthropic · Docker · GitHub Actions — github.com/venkatalakshmiparimala/AgentProbe

- 3-agent LangGraph runtime with tool calling, multi-provider LLM routing, 15-endpoint FastAPI backend, CI/CD + continuous integration via GitHub Actions, React observability dashboard; **5-metric LLM-as-judge eval engine** blocking failing runs before production — machine learning evaluation infrastructure for production agentic systems.

AI on FHIR: HIPAA-Compliant Medical Records NLP Platform
HEALTHCARE

Python · Flask · React · FHIR APIs · BERT · LLMs · Embeddings · Docker — github.com/venkatalakshmiparimala/AI_on_FHIR

- ~60% reduction in clinical retrieval time — NLP platform translating natural-language queries (EN/ES) into FHIR API calls via LLMs + BERT; HIPAA-compliant RBAC, end-to-end implementation from design to deployment.

EDUCATION

M.S. Data Analytics Engineering — George Mason University
May 2026 · GPA 3.87

ML systems, statistical modeling, data engineering, AI pipelines, system design, data governance

B.Tech Computer Science & Engineering — JNTU Anantapur
2013 – 2017

TECHNICAL SKILLS

Languages & Backend: Python, JavaScript, TypeScript, SQL, PL/SQL, R, PHP; FastAPI, Flask, REST APIs, webhooks, API integrations, RabbitMQ, event-driven distributed systems

AI & Agents: LangGraph, LangChain, Agentic AI, Multi-Agent Orchestration, RAG Pipelines, ChromaDB, LLM-as-Judge, LLM Evaluation, machine learning, Agent Observability, OpenAI API, Anthropic API, PyTorch, BERT, FHIR, NLP, YOLO

Infrastructure & Cloud: AWS (S3, EC2, RDS, Glue), Docker, Microservices, CI/CD, continuous integration, GitHub Actions, system design, scalable architecture, design patterns, PostgreSQL, MySQL, HIPAA, HUD, LIHTC, Git, Linux